

User Documentation

Algorithmic Trading System

Table of Contents

- [Introduction](#)
- [Getting Started](#)
- [User Interface](#)
- [Trading Strategies](#)
- [Order Management](#)
- [AI Assistant](#)
- [Analytics and Reports](#)
- [Options Trading](#)
- [Security and Compliance](#)
- [Troubleshooting](#)
- [FAQs](#)
- [Glossary](#)
- [Contact Information](#)

Introduction

Welcome to the Algorithmic Trading System, a cutting-edge platform designed to empower traders by automating and optimizing trading strategies across various financial markets. This documentation serves as your comprehensive guide to understanding and utilizing the system effectively, whether you're a beginner or an experienced quant.

What is the Algorithmic Trading System?

The Algorithmic Trading System is an enterprise-grade platform that provides tools to automate trading strategies for multiple asset classes, including stocks, ETFs, futures, options, forex, and cryptocurrencies. It integrates advanced

artificial intelligence (AI) and machine learning capabilities to assist with strategy development, backtesting, and execution. Key features include:

- A no-code strategy builder for users without programming experience.
- A Python studio for advanced users to code custom strategies.
- An AI Assistant for natural language interaction and task automation.
- Multi-platform access via web and mobile devices.
- Robust security and compliance features to protect your data and transactions.

Key Benefits

- **User-Friendly:** Intuitive interface with customizable dashboards.
- **AI-Enhanced:** Leverage AI for strategy generation, market analysis, and decision-making.
- **Flexible:** Supports both novice traders and advanced quants with no-code and code-based options.
- **Secure:** Built on a zero-trust architecture with enterprise-grade security measures.

This documentation will guide you through every step of using the system, from setup to advanced trading techniques.

Getting Started

Before diving into trading, ensure you have the necessary setup and account configuration completed.

System Requirements

To ensure optimal performance, your device should meet the following specifications:

Requirement	Minimum	Recommended
Browser	Chrome 90+, Firefox 85+	Latest versions
Internet Speed	10 Mbps	50 Mbps

Requirement	Minimum	Recommended
RAM	8 GB	16 GB

Registering for an Account

1. Visit the system's website at [insert system URL].
2. Click the "**Register**" button on the homepage.
3. Fill in your personal details, including name, email, and a secure password.
4. Submit the form and check your email for a verification link.
5. Click the link to verify your account.

Logging In

1. Go to the login page at [insert system URL]/login.
2. Enter your registered email and password.
3. If multi-factor authentication (MFA) is enabled, input the code from your authenticator app.
4. Click "**Log In**" to access the dashboard.

Note: We strongly recommend enabling MFA for enhanced security.

Initial Configuration

After logging in, configure your account to start trading:

1. Connect to a Broker:

- Navigate to "**Settings**" > "**Brokers**".
- Select your preferred broker from the list (e.g., Interactive Brokers, Alpaca).
- Enter your broker's API keys or login credentials as prompted.
- Save the connection.

2. Set Up Data Sources:

- Go to "**Settings**" > "**Data Sources**".
- Choose data providers for paper and live trading (e.g., Yahoo Finance, Alpha Vantage).

- Test the connection to ensure data feeds are active.

Once configured, you're ready to explore the system!

User Interface

The user interface is designed to be intuitive and customizable, providing quick access to all features.

Dashboard

The dashboard is your central hub, displaying key information at a glance:

- **Portfolio Summary:** Current balance, gains/losses, and asset allocation.
- **Recent Trades:** A log of your latest transactions.
- **Market News:** Real-time updates from integrated feeds.
- **Custom Widgets:** Add watchlists, charts, or performance metrics.

Navigation Menu

The left sidebar provides access to core features:

- **Home:** Return to the dashboard.
- **Strategies:** Create and manage trading strategies.
- **Orders:** View and manage active and historical orders.
- **Analytics:** Access market data and performance reports.
- **AI Assistant:** Open the AI interaction panel.
- **Settings:** Adjust account preferences and configurations.

Customizing the Dashboard

Personalize your workspace:

1. Click "**Edit Dashboard**" in the top-right corner.
2. Drag and drop widgets (e.g., charts, news feeds) to your desired layout.
3. Resize or remove widgets as needed.
4. Click "**Save**" to preserve your layout for future sessions.

Chatbot Interface

The AI Assistant's chatbot is located in the bottom-right corner of the screen. Click the icon to open a text or voice input field for interacting with the AI.

Trading Strategies

The system supports creating, testing, and deploying trading strategies using both no-code and code-based tools.

No-Code Strategy Builder

Perfect for users without coding experience:

1. Go to **"Strategies"** > **"New Strategy"** > **"No-Code Builder"**.
2. Drag indicators (e.g., Moving Average, RSI) from the left panel to the canvas.
3. Set parameters (e.g., period = 50 for a moving average).
4. Define conditions and actions (e.g., "Buy when 50-day MA crosses above 200-day MA").
5. Click **"Save"** and name your strategy.

[Insert screenshot of no-code builder interface]

Python Studio

For advanced users who prefer coding:

1. Navigate to **"Strategies"** > **"New Strategy"** > **"Python Studio"**.
2. Access a cloud-hosted Jupyter notebook environment.
3. Write your strategy using Python and pre-installed libraries (e.g., TA-Lib, pandas-ta).
4. Test your code with sample data.

Example: Simple Moving Average Crossover Strategy

```
import pandas as pd
from quantconnect.algorithm import QCAAlgorithm

class SMACrossover(QCAAlgorithm):
```

```
def Initialize(self):
    self.SetStartDate(2020, 1, 1)
    self.SetEndDate(2021, 1, 1)
    self.SetCash(100000)
    self.AddEquity("SPY", Resolution.Daily)
    self.fast = self.SMA("SPY", 50)
    self.slow = self.SMA("SPY", 200)

def OnData(self, data):
    if self.fast.Current.Value > self.slow.Current.Value:
        self.SetHoldings("SPY", 1.0)
    else:
        self.SetHoldings("SPY", 0.0)
```

Backtesting

Evaluate your strategy's performance:

1. Select a strategy from the "**Strategies**" menu.
2. Click "**Backtest**".
3. Set the date range, initial capital, and trading mode (paper or live).
4. Run the simulation.
5. Review results, including:
 - Equity curve
 - Sharpe Ratio
 - Maximum drawdown
 - Total return

Tip: Optimize your strategy by adjusting parameters based on backtest results (see [Analytics and Reports](#)).

Strategy Marketplace

Explore and share strategies:

- **Browse:** View community-shared strategies under "**Strategy Marketplace**".
- **Download:** Select a strategy and click "**Add to My Strategies**".
- **Upload:** Share your own strategies by uploading them to the marketplace.

Order Management

Efficiently manage your trades within the system.

Placing Orders

1. Go to **"Orders"** > **"New Order"**.
2. Select the asset (e.g., "AAPL" for Apple stock).
3. Choose the order type:
 - Market: Execute at the current price.
 - Limit: Set a specific price.
 - Stop: Trigger at a predefined price.
4. Enter the quantity and review the order.
5. Click **"Submit"**.

Monitoring Orders

- View all orders under **"Orders"** > **"Active Orders"**.
- Filter by status (pending, executed, canceled), date, or asset.

Canceling or Modifying Orders

1. Select an active order from the list.
2. Click **"Cancel"** to stop it or **"Modify"** to adjust price/quantity.
3. Confirm the changes.

Order History

- Access past orders under **"Orders"** > **"History"**.
- Export logs as CSV for external analysis.

AI Assistant

The AI Assistant enhances your experience with natural language interaction.

How to Use

- Open the chatbot in the bottom-right corner.
- Type or speak queries/commands, such as:
 - **Query:** "What's the current price of Bitcoin?"
 - **Command:** "Place a market order to buy 100 shares of AAPL."
 - **Analysis:** "Summarize this earnings report."

Uploading Documents

Analyze documents with the AI:

1. Click "**Upload Document**" in the chatbot panel.
2. Select a file (PDF, DOCX, etc.) from your device.
3. Once uploaded, ask questions like:
 - "What are the key points in this report?"
 - "Extract financial data from this document."

Best Practices:

- Use specific queries for accurate responses.
- Ensure files are in supported formats (PDF, DOCX, TXT).

Analytics and Reports

Gain actionable insights with advanced analytics tools.

Market Data

- Access real-time and historical data under "**Analytics**" > "**Market Data**".
- Visualize trends with customizable charts and technical indicators.

Market Screener

Filter assets dynamically:

- Go to "**Analytics**" > "**Market Screener**".
- Use natural language filters, e.g., "Show tech stocks with P/E ratio < 20 and market cap > \$10B."
- Save filters for reuse.

Opportunity Detection

- View AI-generated alerts under **"Analytics" > "Opportunities"**.
- Customize alert criteria (e.g., price spikes, volume surges).

Performance Reports

- Generate reports under **"Analytics" > "Reports"**.
- Metrics include:
 - Total return
 - Volatility
 - Sharpe Ratio
 - Win/loss ratio
- Export reports as PDF or CSV.

Options Trading

Trade options with specialized tools and features.

Overview

Options are contracts giving the right, but not the obligation, to buy or sell an asset at a set price before an expiration date. The system supports both basic and complex options strategies.

Creating Strategies

1. Navigate to **"Strategies" > "New Strategy" > "Options Strategy"**.
2. Choose a template (e.g., covered call, iron condor).
3. Customize legs, strike prices, and expiration dates.
4. Save your strategy.

Tools

- **Greeks Calculator:** Assess risk with Delta, Gamma, Theta, and Vega.
- **Payoff Diagrams:** Visualize potential profit/loss scenarios.
- **Volatility Suite:** Analyze implied and historical volatility surfaces.

Tutorial: Setting Up a Covered Call

1. Select a stock you own (e.g., "AAPL").
2. Choose an expiration date and strike price for the call option.
3. Use the payoff diagram to evaluate outcomes.
4. Place the order to sell the call.

[Insert screenshot of options strategy builder]

Security and Compliance

Protect your account and adhere to regulations.

Best Practices

- **Enable MFA:** Go to "**Settings**" > "**Security**" and activate multi-factor authentication.
- **Strong Passwords:** Use unique, complex passwords and update them regularly.
- **Monitor Activity:** Review audit logs under "**Settings**" > "**Activity**".

Compliance

- **Automated Reports:** Generate compliance documentation under "**Reports**" > "**Compliance**".
- **Regulatory Adherence:** Ensure strategies comply with local financial laws.

Warning: Always consult a financial advisor or compliance officer if unsure about regulations.

Troubleshooting

Resolve common issues with these solutions.

Login Problems

- **Issue:** Can't log in.

- **Solution:** Verify credentials, reset password, or check MFA setup.

Data Feed Issues

- **Issue:** Market data not updating.
- **Solution:** Check internet connection or switch data sources in "**Settings**".

Order Delays

- **Issue:** Orders not executing promptly.
- **Solution:** Verify broker connectivity or check system alerts.

Reporting Issues

- Submit a ticket via "**Help**" > "**Contact Support**" if problems persist.

FAQs

- **How do I switch between paper and live trading?**
 - Go to "**Settings**" > "**Trading Mode**" and toggle the option.
- **Can I use the AI Assistant on mobile?**
 - Yes, the web app is fully responsive on mobile devices.
- **What data sources are available?**
 - Yahoo Finance, Alpha Vantage, Interactive Brokers, and more.

Glossary

- **Agentic RAG:** A retrieval-augmented generation system enhancing AI responses.
- **Zero-Trust Architecture:** A security model requiring continuous verification.
- **Greeks:** Options metrics (e.g., Delta, Gamma) for risk assessment.

Contact Information

For assistance, reach out to our support team:

- **Email:** support@trading-system.com
- **Phone:** 123-456-7890
- **Ticket System:** [Insert link to ticketing system]

This User Documentation is your complete resource for mastering the Algorithmic Trading System. For further help, contact our support team anytime.